

GWENNAP HEAD, United Kingdom

WMO: 038060

Lat: 50.0369N

Lon: 5.6805W

Elev: 207

StdP: 14.59

Time Zone: 0.00 (EUW)

Period: 82-92

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	32.1	34.6	25.6	19.8	35.6	27.8	21.9	37.1	59.1	42.1	52.3	43.4	19.6	60	0.835

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.8	69.8	62.2	67.7	61.9	65.8	61.0	64.2	67.1	63.0	65.9	62.1	64.8	11.3	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	86.7	65.0	61.9	83.5	64.3	60.9	80.5	63.6	29.4	67.2	28.6	66.3	28.0	64.9	71.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 45.6	41.2	35.3	DB	29.2	74.1	4.4	4.3	26.0	77.2	23.4	79.7	20.9	82.1	17.7	85.3	(4)
(5)			WB	27.0	66.1	4.0	2.3	24.1	67.8	21.8	69.2	19.5	70.4	16.6	72.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.9	45.9	43.9	45.9	47.4	51.9	56.0	60.4	60.5	58.0	54.4	49.3	47.9
	DBStd	6.74	4.54	5.24	3.00	3.42	3.68	3.15	3.50	3.21	2.90	2.83	4.16	3.71
	HDD50	685	133	172	127	90	17	0	0	0	0	4	61	81
	HDD65	4811	594	590	591	527	405	271	149	142	211	330	471	530
	CDD50	1359	4	2	1	13	76	179	323	327	239	139	40	16
	CDD65	13	0	0	0	0	0	1	8	4	0	0	0	0
	CDH74	13	0	0	0	0	0	0	11	2	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	18.0	22.3	21.9	19.6	17.3	16.1	14.4	14.3	13.5	15.2	19.2	20.6	21.6
Precipitation	PrecAvg	39.4	4.6	3.5	2.5	2.6	2.3	2.4	2.7	3.0	2.7	3.9	4.9	4.4
	PrecMax	49.2	7.8	7.4	5.6	6.1	5.5	5.4	8.4	7.5	6.7	6.6	9.8	9.3
	PrecMin	31.8	1.0	0.6	0.8	0.5	0.3	0.5	0.3	0.7	0.3	1.4	1.7	1.3
	PrecStd	5.3	1.9	1.9	1.2	1.5	1.2	1.5	1.7	1.7	1.5	1.4	1.9	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.1	53.1	52.9	60.9	68.1	70.5	75.4	72.1	68.4	61.4	56.7	55.6
		MCWB	51.8	50.8	49.9	52.7	58.8	60.5	64.9	63.7	62.6	58.6	55.2	53.7
	2%	DB	52.3	52.0	51.3	56.9	63.9	66.2	70.8	68.8	65.5	60.0	55.8	54.2
		MCWB	50.9	50.3	49.2	50.8	56.6	58.7	62.9	62.7	61.4	57.5	54.4	52.9
	5%	DB	51.7	51.1	50.4	54.2	60.2	63.1	68.3	66.8	63.7	58.8	55.2	53.3
		MCWB	50.3	49.6	48.8	50.2	53.9	57.8	62.4	62.2	60.6	56.6	53.9	52.1
	10%	DB	51.0	50.3	49.7	52.0	57.1	60.7	66.0	65.3	62.3	58.0	54.5	52.5
		MCWB	49.5	49.0	48.2	49.2	52.4	56.9	61.0	61.6	59.6	56.0	53.0	51.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.8	51.9	51.1	53.9	59.7	61.9	67.2	65.4	63.9	60.2	55.9	54.4
		MCDB	52.9	52.2	51.9	58.3	66.3	68.1	71.3	69.9	65.6	60.8	56.1	54.8
	2%	WB	51.7	51.0	50.2	52.1	56.9	59.8	64.3	64.2	62.6	58.5	55.2	53.5
		MCDB	52.0	51.4	50.8	54.7	63.0	63.7	68.5	66.6	64.4	59.0	55.5	54.0
	5%	WB	51.0	50.2	49.5	51.0	54.9	58.7	62.9	63.3	61.5	57.5	54.5	52.6
		MCDB	51.4	50.7	50.1	53.0	59.2	61.7	66.9	65.6	63.0	58.2	54.9	53.1
	10%	WB	50.1	49.3	48.8	50.1	53.3	57.6	61.8	62.4	60.4	56.7	53.7	51.7
		MCDB	50.8	50.0	49.5	51.7	55.9	60.0	65.4	64.7	61.7	57.6	54.3	52.3
Mean Daily Temperature Range	5% DB	MDBR	5.1	5.3	5.0	6.8	6.6	6.2	6.7	5.8	5.9	4.7	5.1	4.6
		MCDBR	4.8	4.5	5.3	10.0	11.5	11.0	10.7	8.3	7.6	5.0	4.1	4.1
		MCWBR	6.2	5.2	5.1	6.9	7.3	7.4	7.2	5.4	6.0	4.9	4.5	5.3
	5% WB	MCDBR	4.8	4.2	4.5	7.4	9.4	8.8	9.0	6.6	6.1	4.3	3.9	4.3
		MCWBR	6.3	5.4	5.0	5.9	6.6	6.4	6.7	5.1	5.7	4.8	4.8	5.5
Clear-Sky Solar Irradiance	taub	0.339	0.352	0.379	0.383	0.391	0.377	0.383	0.367	0.364	0.360	0.337	0.332	
	taud	2.357	2.382	2.331	2.357	2.381	2.451	2.441	2.501	2.491	2.453	2.403	2.320	
	Ebn at Noon	218	247	260	273	275	280	276	276	264	242	217	200	
	Edh at Noon	22	27	33	36	36	34	34	30	28	24	20	19	
All-Sky Solar Radiation	RadAvg	269	519	885	1441	1735	1900	1746	1502	1140	655	345	216	
	RadStd	23	49	97	94	128	174	119	103	78	40	26	23	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page